UNIVERSITY OF CAMBRIDGE

LLM fundamental Agent: Guest Lecture 2

Tutorial

Dr Fabrizio Coiai

Building a basic fundamental Agent

- Recent literature focus on the use of LLM in finance with particular focus on agent based

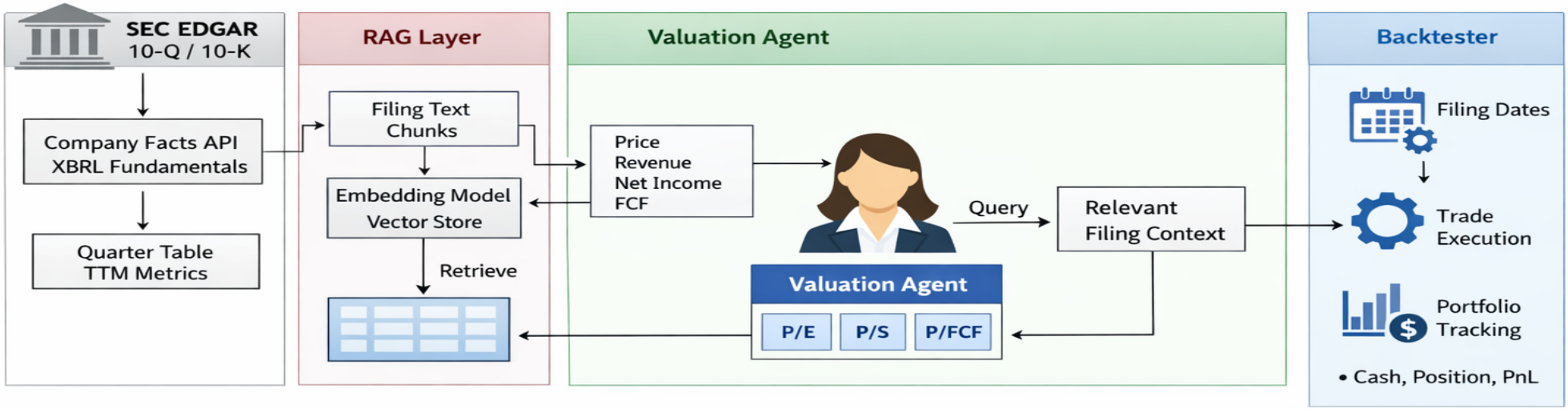
- We took inspiration from the following to develop a basic fundamental equity analyst

- AlphaAgents: Large Language model based Multi-Agents for Equity Portfolio Construction https://arxiv.org/abs/2508.11152

The project

- Uses SEC quarterly fundamentals data plus market prices from yahoo. All data are public available
- From SEC data: revenue, income, cash flow, shares.
- Build trailing 12 months multiple (TTM) valuation metrics
- Build a fundamental valuation agent using Gemini model. Model output:
  1. Buy/Sell/Hold Signal
  2. Confidence and thesis
- Back testing using yahoo price data and signal from the agent

Architecture Diagram



Edgar repository data of annual (10-K) and quarterly (10-Q) reports

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Apple Inc. AAPL on Nasdaq

Investor Toolkit: On

[+] Company Information

Latest Filings (excluding insider transactions)

- February 24, 2026 - 8-K: Current report Filing
  - **5.07** - Submission of Matters to a Vote of Security Holders
  - **9.01** - Financial Statements and Exhibits
- January 30, 2026 - 10-Q: Quarterly report for quarter ending December 27, 2025 Filing
- January 29, 2026 - 8-K: Current report Filing **Earnings release**
  - **2.02** - Results of Operations and Financial Condition
  - **9.01** - Financial Statements and Exhibits
- January 22, 2026 - PX14A6G: Notice of exempt solicitation submitted by non-management Filing
- January 8, 2026 - DEFA14A: Additional definitive proxy soliciting materials and Rule 14(a)(12) material Filing

View filings

Selected Filings

[+] 8-K (current reports)

[+] 10-K (annual reports) and 10-Q (quarterly reports)

[+] Proxy (annual meeting) and information statements

[+] Ownership disclosures

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-Q

(Mark One)

☒ QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the quarterly period ended December 27, 2025
or
☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the transition period from _____ to _____.
Commission File Number: 001-36743

Apple Inc.

(Exact name of Registrant as specified in its charter)

California	94-2404110
(State or other jurisdiction of incorporation or organization)	(I.R.S. Employer Identification No.)
One Apple Park Way Cupertino, California	95014
(Address of principal executive offices)	(Zip Code)

CONDENSED CONSOLIDATED STATEMENTS OF OPERATIONS (Unaudited)
(In millions, except number of shares, which are reflected in thousands, and per-share amounts)

	Three Months Ended	
	December 27, 2025	December 28, 2024
Net sales:		
Products	$ 113,743	$ 97,960
Services	30,013	26,340
Total net sales	143,756	124,300
Cost of sales:		
Products	67,478	59,447
Services	7,047	6,578
Total cost of sales	74,525	66,025
Gross margin	69,231	58,275
Operating expenses:		
Research and development	10,887	8,268
Selling, general and administrative	7,492	7,175
Total operating expenses	18,379	15,443
Operating income	50,852	42,832
Other income/(expense), net	150	(248)
Income before provision for income taxes	51,002	42,584
Provision for income taxes	8,905	6,254
Net income	$ 42,097	$ 36,330
Earnings per share:		
	$ 2.85	$ 2.41
Diluted	$ 2.84	$ 2.40

Valuation agent

```python
        if getattr(self.cfg, "use_llm", True) is False:
            action = "buy" if prior > 0.25 else ("sell" if prior < -0.25 else "hold")
            return ValuationDecision(
                ticker=vin.ticker,
                asof=vin.asof,
                action=action,
                confidence=0.55,
                score=prior,
                thesis="LLM disabled: using rule-based valuation prior only.",
                key_points=[
                    f"P/S={metrics.get('ps')}",
                    f"P/E={metrics.get('pe')}",
                    f"P/FCF={metrics.get('p_fcf')}",
                ],
                risks=["LLM disabled; qualitative context missing."],
                metrics=metrics,
            )

        prompt = f"""
You are a valuation-focused equity analyst.

Ticker: {vin.ticker}
As-of date: {vin.asof.date()}
Price: {vin.price:.2f}
Horizon: {self.cfg.horizon}

Point-in-time valuation metrics:
- Market Cap: {metrics.get("market_cap")}
- TTM Revenue: {metrics.get("ttm_revenue")}
- TTM Operating Income: {metrics.get("ttm_op_income")}
- TTM Net Income: {metrics.get("ttm_net_income")}
- TTM Free Cash Flow: {metrics.get("ttm_fcf")}
- Shares Outstanding: {metrics.get("shares_outstanding")}
- P/S: {metrics.get("ps")}
- P/E: {metrics.get("pe")}
- P/FCF: {metrics.get("p_fcf")}

Rule-based valuation prior (range -1..+1): {prior}

Retrieved filing context:
{filing_context}

Task:
Assess whether the stock is undervalued, fairly valued, or overvalued based on valuation,
cash generation, profitability, and management commentary. Be conservative.

Output exactly:
ACTION: <buy|sell|hold>
CONFIDENCE: <0..1 float>
SCORE: <-1..+1 float>
THESIS: <1-3 sentences>
POINTS: <semicolon-separated>
RISKS: <semicolon-separated>
""".strip()
```

```
SYSTEM
You are a professional equity analyst.

USER
Evaluate the valuation of the following stock.

Ticker: AAPL

Price: 273
Market Cap: 4.1T

TTM Revenue: 383B
TTM Net Income: 96B
TTM Free Cash Flow: 110B

Valuation metrics:
P/S = 10.7
P/E = 28.5
P/FCF = 37

Retrieved filing context:
- Services growth continues to accelerate
- Margins improving
- Continued share buybacks

Based on these metrics and context,
decide whether the stock is:

BUY
SELL
HOLD

Explain reasoning and provide confidence.
```

LLM returns:

```
{
 action: "hold",
 score: 0.2,
 confidence: 0.68,
 thesis: "Valuation elevated relative to growth..."
}
```

Improve LLM with Retrieval-Augmented Generation (RAG)

- Without RAG LLM would rely only on price history or internal memory

- RAG act an external memory and help in limiting ‘hallucination’

- The use of RAG allow the agent to retrieve relevant data (in this case reporting) . At decision time, the agent queries for relevant data before the decision timestamp.

Results and limitations

```
Using tag: RevenueFromContractWithCustomerExcludingAssessedTax
Using tag: EntityCommonStockSharesOutstanding
Adding one small filing snippet to RAG...
RAG document added.
```

date	price	action	position	portfolio_value	decision_score	decision_confidence
2025-12-10	278.779999	buy	60.0	105548.782948	1.0	0.55
2025-12-11	278.029999	buy	60.0	105503.782948	1.0	0.55
2025-12-12	278.279999	buy	60.0	105518.782948	1.0	0.55
2025-12-15	274.109985	buy	60.0	105268.582143	1.0	0.55
2025-12-16	274.609985	buy	60.0	105298.582143	1.0	0.55
2025-12-17	271.839996	buy	60.0	105132.382802	1.0	0.55
2025-12-18	272.190002	buy	60.0	105153.383168	1.0	0.55
2025-12-19	273.670013	buy	60.0	105242.183827	1.0	0.55
2025-12-22	270.970001	buy	60.0	105080.183095	1.0	0.55
2025-12-23	272.359985	buy	60.0	105163.582143	1.0	0.55
2025-12-24	273.809998	buy	60.0	105250.582875	1.0	0.55
2025-12-26	273.399994	buy	60.0	105225.982655	1.0	0.55
2025-12-29	273.760010	buy	60.0	105247.583608	1.0	0.55
2025-12-30	273.079987	buy	60.0	105206.782216	1.0	0.55
2025-12-31	271.859985	buy	60.0	105133.582143	1.0	0.55

```
Cumulative return: 0.051335821427917594

Last thesis:
 LLM disabled: using rule-based valuation prior only.
```

- LLM disable to save quota
- Small manual RAG snippet

```
agent = ValuationAgent(
    llm=llm,
    config=ValuationAgentConfig(
        use_llm=False,  # use True for LMM
        cheap_threshold_ps=3.0,
        expensive_threshold_ps=8.0,
        cheap_threshold_pe=15.0,
        expensive_threshold_pe=30.0,
        cheap_threshold_pfcf=12.0,
        expensive_threshold_pfcf=25.0,
    ),
    filing_rag=filing_rag,
)
```

```
print("Adding one small filing snippet to RAG...")
filing_rag.add_document(
    llm,
    doc_id="aapl_test_doc",
    ticker=ticker,
    filed=pd.Timestamp("2025-05-02"),
    source="10-Q",
    text="Management discussed services growth, gross margin resilience, and ongoing capital returns."
)
print("RAG document added.")
```

Next steps

- The basic project is close to research-grade (full filings and LLM need to be activate), but several improvement must be made. For example,

1. Run on multi-stock universe (S&P500 for example)
2. Convert signal into weights using some form of optimization
3. Improve the agent by adding sentiment for example
4. Adding further factors to the analysis
5. Improve the back testing and use different models (Gemini, OpenAI, Anthropic)
6. This is a relative new field, and you are limited only by your imagination…

Disclosure

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